Introduction to Java Programming

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JDK SE

Java Development Kit Standard Edition.

JDK SE 26 ← current version of Java

IDE (Integrated Development Environment)

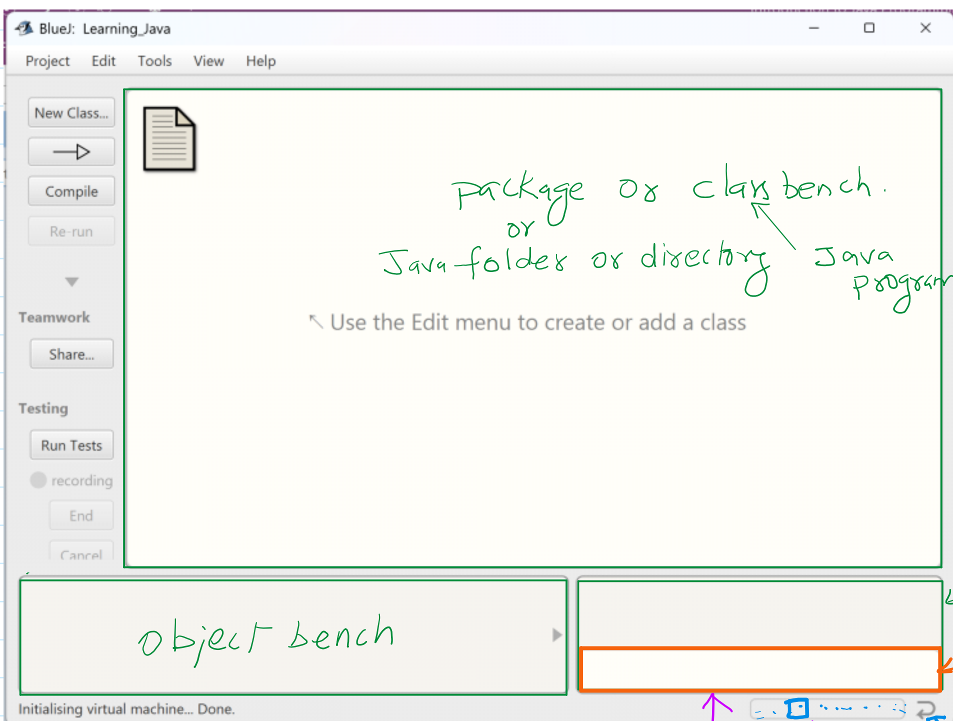
$\boxed{\text{JDK SE}}$

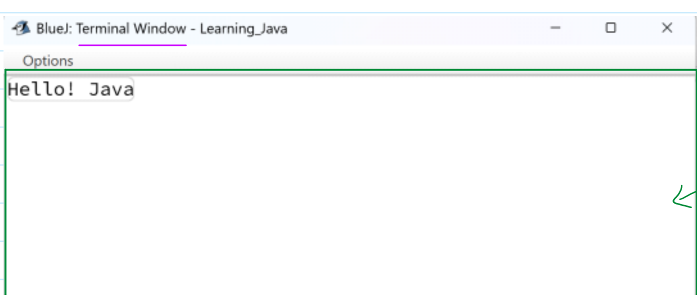
IntelliJ idea for Student Community edition is recomended.

Eclipse

Netbeans Apache foundation.

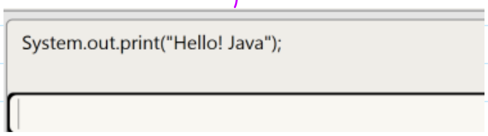
In kids the famous IDE for Java is "BlueJ"





in (data input)

output



What is Java?
Java is a high level object oriented, secure and platform independent programming language widely used for developing:

- Desktop application.
- Web applications.
- Mobile applications.
- Enterprise software.
- Scientific applications.
- Embedded systems.
- Cloud based systems.
- AI models like Claude, etc.

Java was developed by Sun Microsystem in 1995 under the leadership of James Gosling. Later Java become part of Oracle Corporation after Oracle acquired Sun Microsystems.

Why Java is Popular?
Java become popular because it follows the principal:
"Write once, run anywhere" (WORA)
This means a Java program written on one operating system can run on another operating system without modification.

Example:
A Java program written on:

- Windows

Can run on:

- Linux
- macOS
- Unix

Provided Java Virtual Machine (JVM) is installed.

Features of Java

1. **Platform Independent:** Java code is converted into **byte code.** Which can run on any machine having JVM.

2. **Object Oriented:** Java follows Object Oriented programming (OP) concepts such as:
   - **i.** Class
   - **ii.** Object
   - **iii.** Inheritance

iv. Polymorphism
v. Encapsulation
vi. Abstraction

3. **Simple:** Java syntax is easy to learn and resembles with C/C++ But removes complex features like:
   - **a.** Pointers.
   - **b.** Operator overloading
   - **c.** multiple inheritance through classes.

4. **Secure:** Java provides security using:
   - **a)** Bytecode verification
   - **b)** JVM sandbox
   - **c)** Automatic memory management

5. **Robust:** Java is reliable because of:
   - **a)** Exception handling
   - **b)** Strong memory management
   - **c)** Garbage collection

6. **Multithreaded:** Java supports executing multiple tasks simultaneously using threads. Example:
   - **1)** Gaming
   - **2)** Video Streaming
   - **3)** Banking System
   - **4)** Reservation System

7. **Distributed:** Java supports distributed computing through networking APIs.

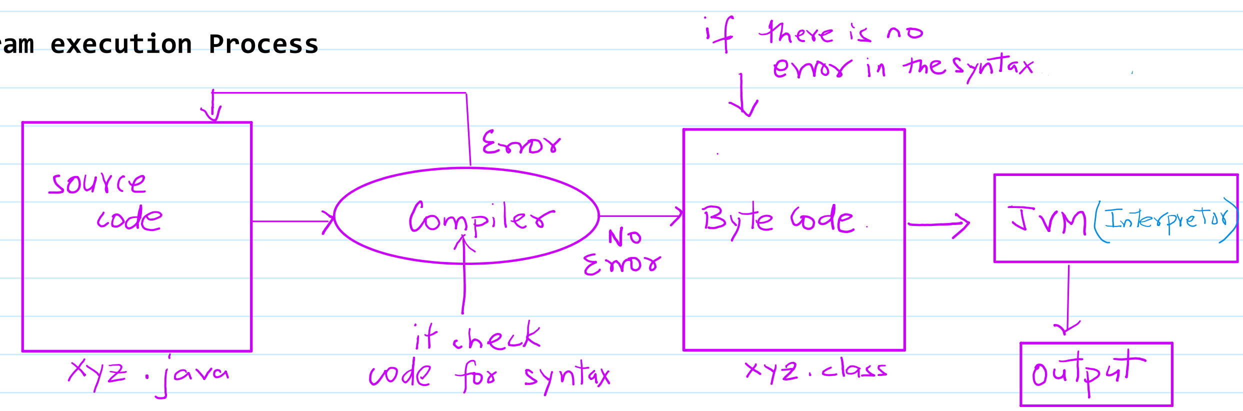
8. **High Performance: Using Just-In-Time(JIT) compiler(Java improves execution speed)**

Java Editions

Java has different additions for different purposes.

Edition	Purpose
Java SE	Standard Edition
Java EE/Jakarta EE	Enterprise Applications
Java ME	Mobile and Embedded Systems

Java program execution Process



Basic Architecture of Java.

Components of Java

1. JVM(Java Virtual Machine): The responsible for executing byte code.

   Functions:

   - □ Loads code
   - □ Verifies code
   - □ Executes program
   - □ Manages memory

2. JRE(Java Runtime Environment): Provides environment to run Java applications.

   Contains:

   - JVM
   - Libraries
   - Supporting files

3. JDK(Java Development Kit): Complete package for Java development.

   Contains:

   - JRE
   - Compiler
   - Debugging tools
   - Documentation tools

Relationship between JVM, JRE and JDK

JDK->contains JRE -> contains -> JVM

Basic Structure of Java Program

```
/**
 * Write a description of class Hello here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Hello
{
    public static void main(String[] args){
        System.out.print("\fHello World!");
    }
}
```

Hello.java.